

Unit 5: Materials, Structures, and Prototype Testing

Quiz 5A: Materials, Loads, Force Types, and Strength-to-Weight

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

1. Strength-to-weight ratio compares...
 - A. Strength to mass or weight
 - B. Color to shape
 - C. Cost to page number
 - D. Time to temperature only
2. Tension is a force that...
 - A. Pulls a material apart
 - B. Pushes a material together
 - C. Twists a material
 - D. Measures color
3. Compression is a force that...
 - A. Pushes a material together
 - B. Pulls a material apart
 - C. Only bends paper
 - D. Changes file names
4. Shear tends to...
 - A. Slide layers or sections past each other
 - B. Only stretch a part
 - C. Only compress air
 - D. Only change color
5. Why is low mass important in aerospace?
 - A. Less mass can reduce energy or lift requirements
 - B. It always makes parts stronger
 - C. It removes safety needs
 - D. It stops testing
6. Material selection should consider...
 - A. Strength, weight, cost, process, and environment
 - B. Only color
 - C. Only what is nearby
 - D. Only student preference
7. A brittle material may...
 - A. Break suddenly with little deformation
 - B. Always stretch forever
 - C. Never fail
 - D. Measure itself
8. A ductile material may...

- A. Deform before breaking
- C. Only shatter

- B. Never bend
- D. Avoid all forces

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|---------------------------------------|---|
| _____ Tension | A. A pulling force. |
| _____ Compression | B. A pushing force. |
| _____ Shear | C. A force where layers or sections slide past each other. |
| _____ Strength-to-weight ratio | D. A comparison of strength to weight or mass. |
| _____ Material property | E. A characteristic such as stiffness, strength, density, or toughness. |

Part C: Short Response

1. Explain why aerospace structures often prioritize strength-to-weight ratio.

2. Give one example of a material property that would matter in a lightweight bracket.

Unit 5: Materials, Structures, and Prototype Testing

Quiz 5B: Structures, Trusses, Spars, and Failure Modes

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- A truss is useful because...
 - Triangles can distribute loads efficiently
 - It always uses more material
 - It removes all loads
 - It only decorates parts
- A spar in an aircraft wing primarily helps...
 - Carry structural loads along the wing
 - Display the logo
 - Hold quizzes
 - Store pencils
- Buckling is a failure often related to...
 - Compression
 - Only color
 - File names
 - Tension only
- A load path describes...
 - How forces travel through a structure
 - Where files are saved
 - The route to the classroom
 - The sketch title
- Why do engineers study failure modes?
 - To design safer and stronger improvements
 - To blame students
 - To avoid redesign
 - To make parts heavier always
- Which shape commonly improves structural stiffness?
 - Triangle
 - Random squiggle
 - Unconnected line
 - Blank space
- A gusset can help...
 - Reinforce a joint
 - Delete a drawing
 - Measure time
 - Export a file
- A structure that is strong but too heavy may be poor for aerospace because...

- A. Mass affects performance
- C. Weight does not matter

- B. Strong parts are never useful
- D. It cannot be drawn

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|---------------------------|--|
| _____ Truss | A. A structure made of connected members, often using triangles. |
| _____ Spar | B. A main structural member in a wing or beam-like system. |
| _____ Buckling | C. A compression-related instability or collapse. |
| _____ Load path | D. The route forces take through a design. |
| _____ Failure mode | E. The way a design breaks, bends, deforms, or stops working. |

Part C: Short Response

1. Explain why triangles are common in lightweight structures.

2. Describe one possible failure mode for a paper or 3D printed tower.

Unit 5: Materials, Structures, and Prototype Testing

Quiz 5C: Test Plans, Data Tables, Failure Analysis, and Redesign

Name: _____ Period: _____

Date: _____ Score: _____

Instructions

Answer each question independently. Choose the best answer for multiple choice, complete the matching section, and respond to short-response prompts in complete sentences.

Score:

/ 25

Part A: Multiple Choice

- A test plan should include...
A. What will be tested, how it will be measured, and success criteria
B. Only team names
C. Only final photos
D. Only a guess
- Why collect data during testing?
A. To support decisions with evidence
B. To fill space only
C. To avoid redesign
D. To replace observations
- Failure analysis asks...
A. What failed, why it failed, and what should change
B. Who to blame first
C. How to hide errors
D. How to skip testing
- A controlled test changes...
A. As few variables as possible at one time
B. Everything randomly
C. Nothing ever
D. Only the page border
- Redesign should be based on...
A. Evidence from testing and observations
B. Only favorite colors
C. Random changes
D. The fastest idea
- Which data table entry is useful?
A. Load supported: 450 g; failure at joint A
B. It broke
C. Bad
D. Cool
- Why test prototypes instead of assuming they work?
A. Testing reveals real performance and failures
B. Assumptions are always correct
C. Testing is only decoration
D. It avoids data

8. What should a team do after a failed test?
- A. Document the failure and plan an evidence-based improvement B. Throw away all notes
- C. Hide the result D. Never test again

Part B: Vocabulary / Matching

Write the letter of the best matching term in the blank.

- | | |
|-------------------------------|---|
| _____ Test plan | A. A plan that defines how performance will be tested and measured. |
| _____ Data | B. Measurements or observations collected during testing. |
| _____ Failure analysis | C. The study of how and why a design failed. |
| _____ Controlled test | D. A test that limits variables so results are meaningful. |
| _____ Redesign | E. Changing a design based on evidence. |

Part C: Short Response

1. Describe what should be included in a good test plan.

2. Explain how failure can be useful in engineering.
